Evaluating NLP Models for Health Misinformation and AI-Generated Text: Performance, Robustness and Shortcut Learning

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***Abstract*—**Natural language processing (NLP) models for health-information classification are commonly evaluated on fixed train–test splits, but high benchmark scores do not necessarily show that they identify medically meaningful evidence of reliability. Performance may instead reflect dataset-specific source, authorship, generator, structural or formatting cues. This concern is increasingly important because AI-generated health misinformation can be fluent and authoritative in style, potentially making unreliable content appear more credible to readers. This study evaluates two dataset-defined text-classification tasks in MM-Health: reliability classification, distinguishing reliable from unreliable human-written health articles, and originality classification, distinguishing human from AI-generated articles. Classical models are compared with BERT, RoBERTa and PubMedBERT. RoBERTa achieves the strongest standard results, with macro-F1 scores of 0.952 for reliability and 1.000 for originality to three decimal places, and makes significantly fewer errors than each alternative in Holm-corrected McNemar tests. Robustness testing nevertheless reveals weaker transfer. A reliability model trained only on human originals falls to 0.614 macro-F1 on one matched AI-generated version per test article, while cross-source scores range from 0.785 to 0.941. Paired error analysis shows that 303 of 1,159 articles are classified correctly in human form but incorrectly after AI generation, with proportionally greater deterioration among unreliable-labelled articles. In originality classification, four held-out generators achieve 0.985–0.998 macro-F1, but held-out Mistral falls to 0.591. Removing a generator-imbalanced leading-Markdown cue leaves the standard result near ceiling at 0.999 but raises held-out Mistral to 0.984. A length-only baseline (0.631), a 17-feature style-only classifier (0.875) and a normalised TF-IDF lexical classifier (0.988) show that substantial structural and lexical separability remains. These findings show that strong benchmark performance may reflect regularities in dataset construction rather than robust medical fact checking or universal AI-text detection.

1. Introduction

Health misinformation can be understood as health-related information that has the potential to mislead health decisions and cause harm. Fridman et al. (2023) argue that harmful health misinformation should be evaluated in terms of its effects on decision-making, including physical, emotional, social and financial harm. They also report that up to 40% of social-media posts contain health misinformation related to vaccinations, eating disorders, treatments and chronic diseases, including cancer. The risk is particularly significant because almost 72% of people use the internet first when seeking health-related information. Reported consequences include delayed treatment, the use of ineffective or potentially harmful therapies, misleading interpretation of evidence, negative effects on mental health, misallocation of health resources and vaccination hesitancy (Borges do Nascimento et al., 2022). Recent advances in generative AI have made health-misinformation detection more challenging because large language models (LLMs) can generate fluent, human-like health content at scale, including content that may appear credible despite being unreliable (Chen and Shu, 2024).

This study evaluates the textual component of MM-Health (Zhang et al., 2025). It examines two dataset-defined tasks: reliability classification, which distinguishes reliable from unreliable health articles, and originality classification, which distinguishes human-written from AI-generated articles. Term frequency–inverse document frequency (TF-IDF) logistic regression and a linear support vector machine (SVM) are compared with BERT, RoBERTa and PubMedBERT. However, the study does not focus only on which model obtains the highest score. Strong benchmark performance does not necessarily show that a model has learned a robust decision rule for the intended task. Misinformation classifiers may rely on shortcuts such as source, writing style, sentiment, length or formatting that correlate with labels within a dataset but do not constitute meaningful evidence of reliability (Wan et al., 2025). Models that depend on these cues may perform well when training and test data share the same construction process, yet degrade when authorship, source or generator changes. This concern is particularly relevant to MM-Health because its human and AI articles differ in production process, vocabulary, structure and formatting.

The study therefore combines benchmark comparison with controlled evaluations under distribution shift and interpretable feature analysis. Reliability is evaluated using matched AI rewrites, a mixed human–AI test set and cross-source testing. Originality is evaluated using cross-generator testing, followed by targeted removal of a leading-Markdown cue. Residual differences are then examined using a length-only baseline, paired stylometric analysis, a 17-feature style-only classifier and a leading-Markdown-normalised TF-IDF lexical classifier based on words and bigrams.

The research questions are:

RQ1. Which of the evaluated classical and transformer models performs best on the official MM-Health splits?

RQ2. Does reliability classification transfer from human originals to matched AI rewrites and unseen source datasets?

RQ3. Does originality classification transfer to an AI generator excluded from training?

RQ4. Which formatting, structural and lexical cues are predictive of MM-Health originality labels?

Reliability is treated as classification according to MM-Health's inherited labels rather than direct medical fact checking. Originality refers only to the human and AI-generator conditions represented in MM-Health, not universal AI-text detection.

2. Related Work

2.1 Models for Misinformation Detection

Health-misinformation detection has used methods ranging from classical machine learning to pretrained transformer models. A systematic review by Schlicht et al. (2023) found

that BERT, RoBERTa and domain-specific transformers had been applied across health-misinformation detection and quality-assessment studies, with pretrained transformers tending to outperform other deep-learning approaches in the reviewed experiments. A more recent scoping review similarly identified transformer models as common approaches for medical-misinformation classification, claim verification and fact-checking, while noting continuing challenges involving contextual understanding, realistic datasets and consistent evaluation (Sun et al., 2025).

Pretrained transformers learn contextual language representations from large text corpora and can be fine-tuned for specific classification tasks. In general fake-news detection, Awalina et al. (2021) reported that BERT outperformed convolutional neural network, bidirectional long short-term memory and hybrid models on an Indonesian dataset. BERT introduced bidirectional transformer pretraining followed by task-specific fine-tuning (Devlin et al., 2019), while RoBERTa modified the BERT pretraining procedure and provided a strong general-domain language model (Liu et al., 2019).

PubMedBERT was pretrained from scratch on biomedical literature using a domain-specific vocabulary and achieved strong performance across biomedical natural language processing tasks (Gu et al., 2022). Its inclusion in this study tests whether biomedical pretraining provides an advantage when classifying health-related news articles. However, domain-specific pretraining is not assumed to be superior because MM-Health contains mixed-source health journalism rather than only biomedical abstracts or clinical text.

2.2 Robustness, Shortcut Learning and AI-Generated Text Detection

Strong performance on a fixed benchmark does not necessarily demonstrate robust generalisation. Geirhos et al. (2020) describe shortcut learning as the use of decision rules that perform well under standard benchmark conditions but fail to transfer when testing conditions change. This concern is directly relevant to misinformation detection. Wan et al. (2025) found substantial performance degradation across misinformation benchmarks when models were exposed to naturally occurring or deliberately introduced shortcut cues, indicating that high benchmark performance may partly depend on superficial dataset correlations rather than transferable evidence.

Distribution shift may also arise from changes in source, domain or textual presentation. Silva et al. (2021) showed that fake-news detection performance can decline on unseen or rarely represented domains, which differ in characteristics such as vocabulary and propagation patterns. Similarly, Wu et al. (2024) found that LLM-induced changes in writing style substantially reduced the performance of existing text-based fake-news detectors, demonstrating sensitivity to stylistic variation even when the underlying veracity label was unchanged. These findings motivate evaluation beyond familiar source and authorship conditions.

Generalisation is also a central problem in AI-generated text detection. Pu et al. (2023) trained detectors on outputs from individual language models and found that none generalised successfully to every unseen generator. Wang et al. (2024) similarly reported difficulties transferring machine-generated-text detectors to unseen generators and domains in the M4 benchmark. Alongside model-based detection, Kumarage et al. (2023) showed that stylometric signals can aid discrimination between human and AI-generated text, supporting the examination of interpretable stylistic characteristics in addition to classifier performance.

These findings motivate the robustness analyses used in this study. MM-Health reliability is evaluated under matched human–AI authorship shift and cross-source conditions, while originality is evaluated using cross-generator testing. Suspected dataset-construction cues are further examined through controlled leading-Markdown normalisation and structural, stylometric and lexical analyses. Together, these experiments assess whether strong standard benchmark results remain stable when authorship, source, generator or superficial textual cues change.

3. Dataset

This study uses the textual component of MM-Health (Zhang et al., 2025), a large-scale health-misinformation benchmark released in 2025. MM-Health was not collected from scratch. Instead, it was assembled using human-written health news articles drawn from four existing open-source misinformation datasets, MM-COVID (Li et al., 2020), MMCoVaR (Chen et al., 2021), Med-MMHL (Sun et al., 2023) and ReCOVery (Zhou et al., 2020). Due to restrictions on access to the Twitter/X API in 2023, the authors excluded all tweet content and retained only news articles. Med-MMHL provided the original text and images directly, whereas the other three datasets mainly provided article URLs. These URLs had to be revalidated and scraped, and links that were no longer accessible were discarded. This process resulted in a total of 5,776 human-written articles. The reliability labels were not newly assigned by the MM-Health authors but were inherited from the original source datasets. Reliability should therefore be understood as a dataset-defined construct rather than a new article-level assessment conducted specifically for MM-Health. For example, ReCOVery assigns credibility labels partly using source-level ratings from organisations such as NewsGuard and Media Bias/Fact Check.

For every human article, five open-source LLMs (Llama-3.1-8B, Qwen2.5-7B, ChatGLM-4-9B, Gemma2-9B and Mistral-v0.3-7B) were used to generate five AI-written versions. Together with the original article, this produced six text variants per item and 34,656 texts in total. The generated versions were not direct paraphrases. GPT-4o first produced a topic summary from the human article, which was then supplied to each open-source LLM to generate a new article. The MM-Health authors also report a human quality evaluation by five trained annotators. Each AI-generated article inherited the reliability label of its corresponding human source article and received an originality label distinguishing AI-generated from human-written content. For the standard experiments, reliability classification uses the 5,776 human originals, whereas originality classification uses all 34,656 variants. The AI-generated articles were generally shorter than the human originals, making document length a potential superficial cue for originality classification.

Both tasks are imbalanced. For reliability, reliable articles outnumber unreliable articles by roughly 4:1, with 4,650 reliable and 1,126 unreliable human-written articles. For originality, the class distribution across all six variants is 1:5 in favour of AI-generated text. These class distributions inform the modelling and evaluation procedures described in the following methodology section.

4. Methodology

4.1 Preprocessing and Article-Grouped Splits

This study adopts MM-Health's official training, validation and test partitions. The partitions are defined at article level, ensuring that the human original and its five AI-generated versions remain in the same partition. This prevents article-level leakage, such as a model encountering the human version of an article during training and an AI-generated version of the same article during testing. The sets of article identifiers (uid) in the training and test partitions were compared directly, with zero overlap found for both tasks. This confirms that no version of the same underlying article appeared in both training and testing.

Reliability classification retained only the human-written originals, producing 4,154 training, 463 validation and 1,159 test articles. Originality classification included the human originals and all five AI-generated versions, producing 24,924 training, 2,778 validation and 6,954 test texts.

Preprocessing was deliberately minimal. Repeated whitespace was collapsed and leading and trailing whitespace were removed, while punctuation, stop words, numbers, Markdown markers and article wording were retained. No stemming, lemmatisation or stop-word removal was applied. This preserved the textual conditions of the benchmark and retained the lexical, structural and formatting features examined in the later shortcut and stylometric analyses. In particular, suspected cues such as leading Markdown were retained in the standard experiments and removed only during the targeted normalisation experiment. No explicit lowercasing was applied by the shared cleaning function, although the TF-IDF vectorizer used its default lowercase conversion.

4.2 Models and Training

Five model configurations were compared on both tasks: TF-IDF logistic regression, a linear SVM, BERT, RoBERTa and PubMedBERT. The classical models represented each article using TF-IDF-weighted unigrams and bigrams. TF-IDF gives greater weight to terms that are frequent within a document but less common across the training corpus, while bigrams retain limited local word-order information. Terms occurring in fewer than two training documents were removed and the vocabulary was capped at 30,000 features. Logistic regression and the linear SVM were trained on these representations using balanced class weights, with logistic regression allowed a maximum of 2,000 optimisation iterations. The same TF-IDF configuration was later reused for the leading-Markdown-normalised lexical experiment. The TF-IDF expression is provided in Appendix A.1.

BERT-base, RoBERTa-base and PubMedBERT were used as the transformer models. Unlike TF-IDF, transformers learn contextual representations in which the representation of each token depends on its surrounding text through self-attention. BERT uses bidirectional contextual encoding, allowing information from both preceding and following tokens to contribute to each representation. RoBERTa retains the BERT architecture but modifies its pretraining procedure, including removal of the next-sentence prediction objective and more extensive pretraining. PubMedBERT follows a BERT-based architecture but was pretrained from scratch on biomedical literature using a domain-specific vocabulary. BERT and RoBERTa therefore provide general-domain transformer baselines, while PubMedBERT tests whether biomedical pretraining provides an advantage for health-news classification.

For classification, each article was tokenised using the tokenizer associated with the corresponding pretrained model and passed through a sequence-classification model. The resulting contextual representation was used to predict either the reliability or originality label, and the pretrained model parameters were fine-tuned on the corresponding MM-Health training data. The executed notebook uses the corresponding tokenizer and sequence-classification implementation for each transformer.

All transformers were fine-tuned using the same fixed configuration: three epochs, batch size 16, learning rate 2×10^{-5}, weight decay 0.01, maximum sequence length 256 and seed 42 unless otherwise reported. These settings were selected as a practical configuration within the available computational budget rather than through extensive hyperparameter tuning. Inputs exceeding 256 tokens were truncated, meaning that the transformer models primarily classified article openings rather than complete articles. Headings, introductory language and early attribution patterns may therefore be disproportionately represented.

For the standard benchmark transformer models, validation macro-F1 was calculated after each epoch, and the checkpoint with the highest value was retained for final evaluation on the held-out test set. The cross-source and cross-generator retraining runs instead used the model obtained after the fixed three training epochs; validation macro-F1 was still evaluated each epoch but was not used to reload a best checkpoint. Fixing the random seed improves reproducibility by controlling pseudorandom processes such as data shuffling and parameter initialisation, although different seeds may still produce slightly different fine-tuning outcomes.

Class imbalance was handled using inverse-frequency class weighting rather than over or under-sampling. Logistic regression and the linear SVM used balanced class weights directly, while transformer class weights were incorporated into the cross-entropy loss so that minority-class errors contributed more strongly during training. Weights were calculated from the relevant training data and recalculated for each cross-source and cross-generator fold because excluding a source or generator changed the training distribution. The class-weight and weighted-loss expressions are provided in Appendix A.2.

4.3 Evaluation Metrics and Statistical Testing

Final performance was evaluated on the held-out test sets using macro-F1 as the primary metric, with accuracy, weighted F1 and per-class precision, recall and F1 also

reported. Macro-F1 calculates F1 separately for each class and gives both classes equal weight. It is therefore appropriate for the imbalanced reliability and originality tasks, where accuracy or weighted F1 may obscure weaker performance on the unreliable or human-written minority class. The F1 and macro-F1 definitions are provided in Appendix A.3.

RoBERTa was compared with TF-IDF logistic regression, the linear SVM, BERT and PubMedBERT using McNemar's test (McNemar, 1947). Because the models classified the same test examples, their prediction correctness was paired. McNemar's test examines whether one model is correct more often than the other on examples where their predictions disagree and therefore tests differences in prediction correctness rather than differences in macro-F1 directly. When fewer than 25 discordant examples were available, an exact two-sided binomial test was used instead of the chi-squared approximation. The test statistic is provided in Appendix A.4.

Four RoBERTa-versus-alternative comparisons were conducted within each task, so the resulting p-values were adjusted using Holm's procedure (Holm, 1979) to control the family-wise error rate. Statistical significance was assessed at 0.05 and interpreted alongside the magnitude of the performance differences and the subsequent robustness results. The Holm procedure is summarised in Appendix A.5.

4.4 Reliability Robustness Evaluation

Reliability robustness was evaluated with RoBERTa under two forms of distribution shift: authorship shift and source shift. For authorship shift, the human-trained RoBERTa model was applied without retraining to one reproducibly selected AI-generated version of each article in the official reliability test set. This produced 1,159 matched human–AI article pairs because the official test partition contains 1,159 human-written articles. Using the same uids in both conditions ensured that the underlying articles, inherited reliability labels, class distribution and number of observations remained unchanged. Only one AI-generated version was retained per article so that articles with five generated variants did not receive greater influence than their human originals. This provides a zero-shot test of whether the learned reliability decision rule transfers when the same underlying article topics and inherited labels are expressed through AI-generated text.

A mixed-authorship condition was also constructed from the same 1,159 test articles, comprising 579 human originals and 580 AI-generated versions. Authorship was allocated within each reliability class so that the original reliability-label distribution was preserved. This provides an additional test of reliability classification when human- and AI-generated articles occur together rather than under a single authorship condition.

A paired prediction-level error analysis was conducted to examine how individual classifications changed after AI generation. Each human article and its matched AI-generated version were compared by uid and categorised as correct in both conditions, correct only in the human condition, correct only in the AI condition, or incorrect in both. Errors on AI-generated articles were further divided into false-reliable predictions, where an unreliable-labelled article was predicted as reliable, and false-unreliable predictions, where a reliable-labelled article was predicted as unreliable. This analysis identifies newly introduced, corrected and persistent errors under authorship shift while retaining the same underlying article and inherited label. It measures disagreement with MM-Health's dataset-defined labels rather than independently assessing medical accuracy.

Source robustness was evaluated separately using cross-source evaluation. In each fold, one of the four MM-Health component datasets was excluded entirely, and a fresh RoBERTa model was trained on human-written articles from the remaining three sources before being tested on the excluded source. A fixed 15% of the remaining development pool was reserved by uid for validation, and class weights were recalculated for each fold. This evaluates whether reliability classification transfers to a source collection absent from training. However, because excluding a source can simultaneously change topic coverage, editorial conventions, class prevalence and labelling procedures, differences between cross-source folds cannot be attributed to a single factor.

4.5 Originality Generalisation and Leading-Markdown Normalisation

Originality robustness was evaluated using cross-generator testing. In the standard originality benchmark, human articles and outputs from all five AI generators are represented across the training, validation and test partitions. This measures performance under familiar generator conditions but does not establish whether the classifier transfers to a generator absent during model development. Cross-generator evaluation therefore excluded one generator at a time from both training and validation and assessed RoBERTa on that unseen generator.

For each cross-generator fold, all human-written articles were retained together with AI-generated articles from the remaining four generators, and a fresh RoBERTa model was fine-tuned using the configuration described in Section 4.2. Class weights were recalculated within each fold because excluding one generator changed the training distribution. The corresponding test set paired each human original with the version produced by the excluded generator. This created a balanced, article-matched comparison in which the human and AI classes covered the same underlying topics, reducing the influence of topic differences and the original 1:5 originality class imbalance. The procedure was repeated with ChatGLM, Gemma, Llama, Mistral and Qwen each serving once as the unseen generator.

ChatGLM, Gemma, Llama and Qwen were evaluated using seed 42. Because the Mistral fold showed greater variation, it was repeated using seeds 42, 43 and 44, with the mean macro-F1 and sample standard deviation reported. Repeating this fold allowed the stability of the observed cross-generator behaviour to be assessed rather than relying on a single training run. The Mistral fold was subsequently used to investigate a potential formatting shortcut. Inspection of article openings identified a generator-imbalanced leading-Markdown pattern, defined as an initial Markdown heading marker or bold-title delimiter. A targeted normalisation removed only this opening formatting while preserving the title words, remaining

article text and Markdown appearing elsewhere in the document.

Normalisation was applied consistently to the training, validation and test partitions before retraining, avoiding an artificial mismatch between development and evaluation data. RoBERTa was first retrained on the normalised official originality split to determine whether the cue was necessary for performance under familiar generator conditions. The normalised cross-generator Mistral experiment was then repeated using seeds 42, 43 and 44 to assess whether removing the formatting cue altered transfer to the unseen generator.

4.6 Residual Shortcut and Stylometric Analysis

Following leading-Markdown normalisation, further analyses examined whether originality remained predictable from document length, writing style and lexical patterns. Three classification models and one paired statistical analysis were used to separate these different sources of information.

First, a length-only logistic regression classifier was trained on the normalised official training split using total word count as its sole predictor and balanced class weights, then evaluated on the corresponding test split. Logistic regression learns a coefficient for article length and converts the resulting linear score into a probability of the human or AI class. With only one predictor available, any discrimination must therefore arise from systematic differences in article length rather than vocabulary or writing style. This provides a deliberately restricted baseline for assessing how much originality information is contained in document length alone.

Second, a paired stylometric analysis examined whether structural and writing-style characteristics differed systematically between human and AI-generated versions of the same articles. Each human article was matched by uid with its five AI-generated versions. For each feature, the five AI values were averaged first, producing one AI value per uid that was then paired with the corresponding human value. The article uid was therefore the unit of analysis, preventing the five related generations from being treated as independent observations. Nineteen structural and stylistic measures were examined, including sentence structure, average word length, lexical diversity, pronoun use and punctuation, with full definitions provided in Appendix B.1. For each feature, the paired difference was calculated as AI minus human. A two-sided Wilcoxon signed-rank test (Wilcoxon, 1945) was then applied to the paired differences. The test ranks the absolute non-zero differences, retains their signs, and assesses whether the paired differences are systematically centred away from zero without requiring a normal distribution. Because this procedure was repeated across 19 features, the resulting p-values were adjusted using the Benjamini–Hochberg procedure (Benjamini and Hochberg, 1995) to control the false-discovery rate across the set of comparisons. Paired Cohen's d_z (Lakens, 2013) was also calculated by dividing the mean AI-minus-human difference by the standard deviation of the paired differences. This provides a standardised measure of effect magnitude in addition to statistical significance, with positive values indicating higher values in AI-generated text and negative values indicating higher values in human text. The corresponding paired-analysis expressions are provided in Appendix A.6.

Third, a style-only logistic regression classifier was trained on the normalised official training split and evaluated on the corresponding test split to determine whether these stylistic characteristics could collectively predict originality. Seventeen features were used, excluding raw word count and sentence count to separate stylistic prediction from the document-length effects examined by the length-only baseline. The features were first standardised using means and standard deviations estimated from the training data, placing measurements with different scales on a comparable basis. Logistic regression then learned one coefficient for each feature and combined them into a single linear decision score used to classify an article as human or AI. Unlike the paired analysis, which tests each characteristic individually, this model assesses whether multiple stylistic signals are jointly useful for prediction. The fitted coefficients were also examined to identify which features were more strongly associated with either class while accounting for the remaining features.

Finally, a TF-IDF logistic regression classifier was trained on the normalised official training split and evaluated on the corresponding test split to examine lexical information after leading-Markdown normalisation. Each article was represented by TF-IDF-weighted unigrams and bigrams, so words and short phrases receive greater weight when they are frequent within an article but less common across the wider training corpus. Logistic regression then learns coefficients for these lexical features and combines them to predict the human or AI class. This differs from the style-only model because the features are derived from the vocabulary itself rather than being predefined stylistic measurements. The same TF-IDF and balanced class-weight settings as the standard lexical baseline were used. For interpretation, reported terms were restricted to those occurring in at least 0.5% of training documents to reduce emphasis on rare and potentially unstable terms. This restriction applied only to coefficient reporting and did not alter the vocabulary used to train the classifier.

5. Results and Discussion

5.1 Standard Benchmarks

Table 1. Standard test macro-F1 by task

Model	Reliability	Originality
TF-IDF + Logistic Regression	0.854	0.988
TF-IDF + Linear SVM	0.901	0.996
BERT	0.918	0.993
PubMedBERT	0.920	0.998
RoBERTa	0.952	1.000

RoBERTa achieved the highest macro-F1 on both tasks, with 0.952 for reliability and 1.000 for originality. For reliability, its class-specific F1 scores were 0.98 for reliable and 0.92 for unreliable articles. Holm-corrected McNemar tests confirmed that RoBERTa made significantly fewer errors than each of the four alternative models on the same test examples in both tasks (all adjusted $p < 0.05$).

RoBERTa was therefore selected for the subsequent robustness experiments.

The classical TF-IDF models also performed strongly, particularly the linear SVM, which achieved macro-F1 scores of 0.901 for reliability and 0.996 for originality. Because these models use unigram and bigram features rather than contextual representations, their performance indicates that substantial label information is recoverable from lexical and short phrase-level patterns. For originality, the near-perfect TF-IDF results show strong lexical separability under the official MM-Health distribution, but do not establish general AI-text detection capability.

PubMedBERT did not outperform RoBERTa despite its biomedical pretraining. This result shows only that biomedical pretraining did not provide the strongest representation for these MM-Health tasks under the evaluated configuration; it should not be interpreted as evidence that biomedical pretraining is generally ineffective.

5.2 Reliability Robustness

The standard reliability benchmark evaluates human-written test articles from source datasets that are also represented in the training data. The matched AI-generated, mixed-authorship and cross-source experiments assess whether RoBERTa maintains its performance when authorship or source conditions change.

Table 2. Human-trained RoBERTa under authorship shift

Test condition	Accuracy	Macro-F1
Human originals	0.971	0.952
Matched AI rewrites	0.720	0.614
Mixed human–AI set	0.837	0.761

When the human-trained RoBERTa classifier was applied zero-shot to matched AI-generated versions of the same test articles, macro-F1 fell from 0.952 to 0.614, a reduction of 0.338. Zero-shot transfer means that the model was applied directly to AI-generated text without retraining or adaptation, despite having been trained only on human-written reliability examples. Because the human and AI conditions used the same matched uids and inherited reliability labels, the decline cannot be attributed to differences in the articles evaluated or to a changed class distribution. Instead, it indicates that the learned reliability decision rule did not transfer robustly from human-written to AI-generated text.

Paired error analysis was used to examine how predictions changed for individual articles across the two authorship conditions. 303 articles (26.14%) changed from a correct prediction on the human-written version to an incorrect prediction on the matched AI-generated version, whereas only 13 (1.12%) changed from incorrect to correct. The overall deterioration therefore reflected a strong imbalance towards previously correct predictions becoming errors after AI generation. The effect was proportionally greater for unreliable-labelled articles: their F1 score fell from 0.921 to 0.411, and 95 of 227 (41.85%) became new errors after AI generation, compared with 208 of 932 reliable-labelled articles (22.32%). Overall, 114 of the 227 unreliable-labelled AI-generated articles (50.22%) were predicted as reliable. This shows that the human-trained classifier was particularly poor at preserving correct predictions for unreliable-labelled articles under authorship shift. In other words, articles that it correctly recognised as unreliable in human-written form were frequently classified as reliable when presented in AI-generated form.

However, the experiment does not establish that the decline was caused specifically by writing style. The MM-Health AI versions were generated through a topic-summary-to-generation process rather than direct paraphrasing, so generation may alter wording, emphasis, detail, attribution and the information appearing within the model's first 256 input tokens. The results therefore demonstrate sensitivity to the change from human-written to AI-generated text, but do not isolate which particular textual changes caused the deterioration.

The mixed human–AI condition produced an intermediate macro-F1 of 0.761, compared with 0.952 for human originals and 0.614 for matched AI-generated versions. Performance therefore remained substantially below the standard human-only benchmark when both authorship conditions were represented, providing further evidence that reliability performance was sensitive to authorship conditions.

Table 3. Cross-source reliability performance

Held-out source	Accuracy	Macro-F1
Med-MMHL	0.963	0.941
ReCOVery	0.962	0.910
MMCoVar	0.950	0.894
MM-COVID	0.827	0.785

Cross-source macro-F1 ranged from 0.785 to 0.941, compared with 0.952 on the standard MM-Health test split, where articles from all four component datasets were represented during training. In the cross-source evaluation, one component dataset was instead excluded entirely from training and used as the test set. Performance was strongest when Med-MMHL and ReCOVery were held out, lower for MMCoVar, and lowest for MM-COVID. Here, source refers to an MM-Health component dataset rather than an individual publisher. Because holding out a component dataset can simultaneously change factors such as topic coverage, temporal context, outlet composition, class prevalence, writing conventions and labelling procedures, the differences between folds cannot be attributed to any single factor. Nevertheless, the results show that reliability performance decreases to varying degrees when the model is tested on a source collection that it did not encounter during training. The lowest cross-source macro-F1 (0.785) remained above the 0.614 obtained on matched AI-generated versions, meaning that the observed deterioration under authorship shift was larger than in any individual cross-source fold. The two experiments are not directly equivalent: the matched comparison holds uids and labels constant while changing the generated text, whereas cross-source evaluation changes several dataset characteristics simultaneously. Together, however, the results show that the strong standard reliability score is not fully robust to changes in either authorship or source distribution.

5.3 Originality Generalisation and the Leading-Markdown Shortcut

The standard originality benchmark includes AI-generated articles from all five MM-Health generators in both training and testing, so it evaluates performance under familiar generator conditions. Cross-generator evaluation provides a stronger test of generalisation by excluding one generator from training and validation and then testing RoBERTa on articles produced by that unseen generator.

Table 4. Originality robustness and leading-Markdown normalisation

Condition	Macro-F1
Standard raw	1.000
Held-out ChatGLM	0.998
Held-out Gemma	0.997
Held-out Llama	0.985
Held-out Qwen	0.996
Held-out Mistral (raw)	0.591 (SD 0.092)
Standard normalised	0.999
Held-out Mistral (normalised)	0.984 (SD 0.015)

Note: ChatGLM, Gemma, Llama and Qwen are seed-42 point estimates. Mistral values are means across seeds 42–44; SDs are sample SDs rounded to three decimals.

RoBERTa generalised strongly to four of the five unseen generators. Held-out ChatGLM, Gemma, Llama and Qwen achieved macro-F1 scores between 0.985 and 0.998. In contrast, held-out Mistral achieved a mean macro-F1 of only 0.591 across seeds 42–44, with an SD of 0.092. The low mean shows that the classifier transferred poorly to Mistral, while the variation across seeds indicates that this failure was also unstable across training runs. Therefore, the near-perfect standard macro-F1 of 1.000 did not guarantee equally strong performance when the model encountered a generator that was absent from training.

Inspection of the generated articles identified a possible explanation for this Mistral-specific failure. A leading-Markdown pattern, consisting of an opening Markdown heading or bold title, occurred in 91.7–99.6% of articles produced by the four generators used to train the Mistral-held-out model, but in only 0.1% of Mistral articles and 2 of 5,776 human articles. Leading Markdown was therefore strongly associated with the AI class during training, while unseen Mistral resembled the human class on this formatting feature. This made leading Markdown a plausible generator-specific shortcut that could reduce transfer to Mistral.

To test this explanation, the leading-Markdown formatting was removed and RoBERTa was retrained. On the standard originality split, macro-F1 changed only slightly from 1.000 to 0.999, showing that the cue was not necessary for near-perfect performance when all generators were represented in training. In contrast, held-out Mistral increased substantially from 0.591 ± 0.092 to 0.984 ± 0.015 after normalisation, an improvement of 0.393 in mean macro-F1. This large recovery provides stronger evidence than the formatting frequencies alone that the leading-Markdown imbalance materially contributed to the poor Mistral transfer.

The normalisation experiment does not show that leading Markdown was the only feature influencing RoBERTa's predictions, since other generator-dependent lexical or structural cues may also be present. However, the results demonstrate that a formatting pattern associated with particular generators can remain largely hidden under the standard benchmark yet strongly affect performance on an unseen generator. Cross-generator evaluation is therefore important for assessing whether high originality-classification performance reflects robust generalisation rather than familiarity with generator-specific production patterns.

5.4 Residual Structural and Lexical Cues

The leading-Markdown intervention largely resolved the Mistral-specific failure, but RoBERTa still achieved 0.999 macro-F1 on the normalised official split. Further analyses therefore examined what structural and lexical patterns continued to distinguish the MM-Health originality labels after normalisation.

Table 5. Originality feature-representation comparison after normalisation

Representation	Information available	Macro-F1
Length only	Article word count	0.631
Style only	17 stylometric features	0.875
TF-IDF lexical	Words and bigrams	0.988
RoBERTa	First 256 contextual tokens	0.999

Length alone provided some predictive signal. After leading-Markdown normalisation, human test articles had a median length of 652 words compared with 472 words for AI-generated articles, reflecting a similar difference in the training data. Using word count as its only predictor, logistic regression learned an approximate decision boundary of 568 words, tending to classify shorter articles as AI-generated. However, it achieved only 0.631 macro-F1, with a human-class F1 of 0.43 compared with 0.83 for AI. This shows that document length was associated with the MM-Health originality labels but could not reliably separate the two classes, as shorter human and longer AI articles substantially overlapped. Length therefore provides a dataset-specific cue, but cannot explain the near-perfect originality performance of the richer models.

The paired stylometric analysis showed that the separation between human and AI-generated articles extended beyond document length. Each human article was compared with the mean of its five corresponding AI-generated versions, so the comparison was made within matched article groups rather than between unrelated texts. Seventeen of the nineteen features remained significant after Benjamini–Hochberg correction, indicating systematic structural and stylistic differences across the matched articles. Paired Cohen's d_z was used to describe the magnitude and direction of these differences, with positive values indicating higher values in AI-generated text and negative values indicating higher values in human-written text. Average word length showed the largest effect (d_z = 2.1363) and was higher in AI-generated text for 98.4% of matched article groups,

indicating a particularly strong and consistent difference. In contrast, human articles showed greater sentence-length variation (d_z = -0.8513), higher stopword use (d_z = -0.6966) and greater document length (d_z = -0.4103). The negative signs indicate that these features were higher in human articles, while the larger absolute effect sizes for sentence-length variation and stopword use indicate stronger differences than for document length. Overall, these results show that MM-Health human and AI texts differed systematically across several aspects of writing style and structure, rather than only in article length. However, these patterns describe MM-Health and should not be interpreted as universal characteristics of human and AI-generated writing. The paired results are summarised in Figure B1 and Table B2 (Appendix B.2).

To assess whether these stylometric differences were useful collectively for prediction, a style-only logistic regression was trained using 17 standardised features, excluding raw word and sentence counts. It achieved a macro-F1 of 0.875, substantially higher than the 0.631 obtained from length alone, while human-class F1 increased from 0.43 to 0.80. This shows that writing-style and structural features provided considerable predictive information beyond document length. The model coefficients identify conditional associations within MM-Health and should not be interpreted causally or as evidence of the features used internally by RoBERTa. The corresponding style-only coefficient plot is provided in Figure B2 (Appendix B.3).

Lexical information provided an even stronger signal. After leading-Markdown normalisation, the TF-IDF logistic regression used only unigram and bigram frequencies yet achieved 0.988 macro-F1, with class-specific F1 scores of 0.98 for human and 1.00 for AI-generated articles. This near-perfect class-balanced performance shows that words and short phrases alone retained enough information to distinguish the MM-Health originality labels under the official split. The result is only 0.011 below RoBERTa's 0.999, indicating that most of the benchmark separability was already recoverable without contextual transformer representations. However, this does not establish that RoBERTa relies on the same lexical features, since the two models use different representations and training procedures.

Overall, the progression from 0.631 macro-F1 using length alone to 0.875 using stylometric features, 0.988 using lexical TF-IDF and 0.999 using RoBERTa shows that document length explains only part of the originality separation. Structural and stylistic information provides substantially greater discrimination, while lexical information alone recovers most of the official-split performance. Removing leading Markdown therefore addressed one generator-specific shortcut without removing the broader separability between human and AI articles in MM-Health. These analyses identify predictive patterns within this dataset rather than universal characteristics of AI-generated text or direct evidence of the features used internally by RoBERTa.

6. Limitations and Future Work

The main limitations arise from MM-Health's construction and the experimental scope. Its AI articles were produced by five open-source generators using the same summary-to-generation pipeline, so other models, prompts or generation procedures may produce different cues. The matched-authorship reliability test used one reproducibly selected AI version per article, while most transformer experiments used one seed; only the anomalous Mistral condition was repeated across seeds 42–44. Cross-source folds also change topic coverage, outlet composition, class prevalence and labelling procedures simultaneously. AI-generated articles inherit the reliability labels of their human sources, which enables matched evaluation but does not guarantee that every generated version preserves all reliability-relevant information. The structural, stylometric and lexical analyses identify predictive associations within MM-Health rather than universal, causal or internal RoBERTa features; the leading-Markdown intervention provides stronger evidence for that specific cue, although other correlated cues may remain. The originality McNemar tests also operate at text level and do not explicitly model dependence among the six variants derived from each article.

Future work should evaluate a wider range of generators, prompts and random seeds; systematically perturb formatting, structural and lexical cues; and test longer contexts than the first 256 tokens used here. For reliability, work could also move beyond predicting inherited MM-Health labels towards evidence-based verification. Retrieval-augmented generation (RAG), for example, could retrieve information from trusted medical sources before claims are assessed, extending beyond the current dataset-defined classification task.

7. Conclusion

This project compared classical and transformer models for MM-Health reliability and originality classification. RoBERTa achieved the strongest standard performance, with macro-F1 of 0.952 for reliability and 1.000 for originality, but the robustness experiments showed that these scores were sensitive to changes in authorship, source and generator conditions. Reliability declined under matched AI-generated and cross-source evaluation, while originality exposed a generator-specific leading-Markdown shortcut that substantially affected transfer to unseen Mistral. After that cue was removed, strong structural and lexical separability remained within MM-Health.

The two tasks therefore support the same conclusion: high performance on a fixed benchmark is not sufficient evidence of robust, transferable learning of the intended concept. Reliability predictions may partly reflect source, authorship or other textual cues and should not be interpreted as direct medical fact checking; originality patterns are predictive within MM-Health but do not establish universal characteristics of AI-generated text. Robustness testing, distribution-shift evaluation and analysis of potential shortcuts should therefore accompany benchmark scores before strong claims are made about model capability or generalisation.

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Appendix A. Mathematical and Statistical Expressions

The expressions below summarise the main mathematical and statistical procedures directly used in the experiments. Standard model architectures are described in Section 4.2 rather than reproduced mathematically here. Symbols are defined immediately below each expression.

A.1 TF-IDF representation

$$\begin{aligned} \mathrm{idf}(t) &= \log\left(\frac{1+N}{1+\mathrm{DF}(t)}\right)+1, \\ v(t,d) &= \mathrm{TF}(t,d)\ \mathrm{idf}(t), \\ \mathbf{x}(d) &= \frac{\mathbf{v}(d)}{\|\mathbf{v}(d)\|_2} \end{aligned} \tag{A1}$$

Here, TF is the raw term count, DF is training-document frequency and N is the number of training documents. The executed notebook used scikit-learn's default smoothed IDF and L2 normalisation, with unigrams and bigrams, min_df = 2 and max_features = 30,000.

A.2 Class weighting and weighted transformer loss

$$w_c = \frac{N}{K\, n_c} \tag{A2}$$

Here, w_c is the weight for class c, N is the total training size, K is the number of classes, and n_c is the number of training examples in class c.

$$L = \frac{\sum_i w_{y_i}\,[-\log p(y_i \mid x_i)]}{\sum_i w_{y_i}} \tag{A3}$$

For training example i, the true-class log-probability is weighted by the corresponding class weight. PyTorch's default mean reduction normalises the weighted losses by the sum of the true-class weights in the batch.

A.3 F1 and macro-F1

$$F1_c = \frac{2P_cR_c}{P_c+R_c} \tag{A4}$$

Pc, Rc and F1c denote class-specific precision, recall and F1, respectively; K is the number of classes.

$$\mathrm{Macro\text{-}F1} = \frac{1}{K}\sum_{c=1}^{K} F1_c \tag{A5}$$

Macro-F1 gives equal weight to each class.

A.4 McNemar test

$$\chi^2 = \frac{(|b-c|-1)^2}{b+c} \tag{A6}$$

b and c are the discordant correctness counts. The exact two-sided binomial test was used when fewer than 25 discordant examples were available.

A.5 Holm-adjusted p-values

$$p_{(i)}^{\mathrm{Holm}} = \min\left\{1, \max_{1\le j\le i}\left[(m-j+1)p_{(j)}\right]\right\} \tag{A7}$$

After the m raw p-values are ordered from smallest to largest, the cumulative maximum is applied, values are clipped at 1, and adjusted p-values are returned to the original comparison order.

A.6 Paired stylometric analysis

$$d_{ij} = \frac{1}{5}\sum_{g=1}^{5} AI_{igj} - H_{ij} \tag{A8}$$

For article i and feature j, the human value is subtracted from the mean of its five AI-generated versions. Positive differences are higher in AI-generated text.

$$d_z = \frac{\bar{d}}{s_d} \tag{A9}$$

The numerator is the mean paired AI-minus-human difference and the denominator is its sample standard deviation.

$$q_{(i)} = \min_{j\ge i}\left\{\min\left(1, \frac{m\,p_{(j)}}{j}\right)\right\} \tag{A10}$$

The m Wilcoxon p-values are ordered before the Benjamini–Hochberg reverse cumulative minimum is applied. Tests were two-sided with zero_method = "wilcox", matching the executed notebook.

A.7 Feature standardisation

$$z = \frac{x-\mu_{\mathrm{train}}}{\sigma_{\mathrm{train}}} \tag{A11}$$

Here, μ_train and σ_train are the feature mean and standard deviation estimated from the training split and reused to transform the test split.

Appendix B. Stylometric Analysis Details

This appendix provides supporting feature definitions and results for Sections 4.6 and 5.4. All analyses use the leading-Markdown-normalised official originality data. The reported patterns describe associations within MM-Health and should not be interpreted as universal characteristics of human or AI writing.

B.1 Feature extraction

Word tokens were identified using alphabetic sequences with optional internal apostrophes. Sentences were split after full stops, question marks or exclamation marks followed by whitespace; segments without a word token were excluded, and the complete article was treated as one sentence when the split produced none. Rates were calculated per 100 words.

Table B1. Stylometric feature definitions

Feature	Notebook-supported definition
Number of words	Count of alphabetic word tokens, allowing an internal apostrophe.
Number of sentences	Number of punctuation-based sentence segments containing at least one word token.
Average sentence length	Mean number of word tokens per sentence.
Sentence-length variation	Population standard deviation of sentence word counts (ddof = 0).
Average word length	Mean number of characters per alphabetic word token.
Lexical diversity (Herdan C)	log(unique lower-case word types) / log(word tokens); set to zero when undefined.
Stopwords per 100 words	Rate of tokens contained in scikit-learn ENGLISH_STOP_WORDS.
First-person pronouns per 100 words	Rate for i, me, my, mine, myself, we, us, our, ours and ourselves.
Second-person pronouns per 100 words	Rate for you, your, yours, yourself and yourselves.
Modal verbs per 100 words	Rate for can, could, may, might, must, shall, should, will and would.
Digit-containing tokens per 100 words	Rate of tokens containing at least one digit.
Commas per 100 words	Comma-character count per 100 words.
Semicolons per 100 words	Semicolon-character count per 100 words.
Colons per 100 words	Colon-character count per 100 words.
Question marks per 100 words	Question-mark count per 100 words.
Exclamation marks per 100 words	Exclamation-mark count per 100 words.
Quotation marks per 100 words	Combined count of straight and left/right double quotation marks per 100 words.
Parentheses per 100 words	Combined count of opening and closing parentheses per 100 words.
Uppercase tokens per 100 words	Rate of alphabetic tokens longer than one character for which isupper() is true.

B.2 Paired within-article analysis

Figure B1 reproduces the notebook plot of the 12 largest absolute paired effect sizes. Positive d_z values indicate higher values in AI-generated text and negative values indicate higher values in human-written text. Table B2 reports d_z for all 19 features.

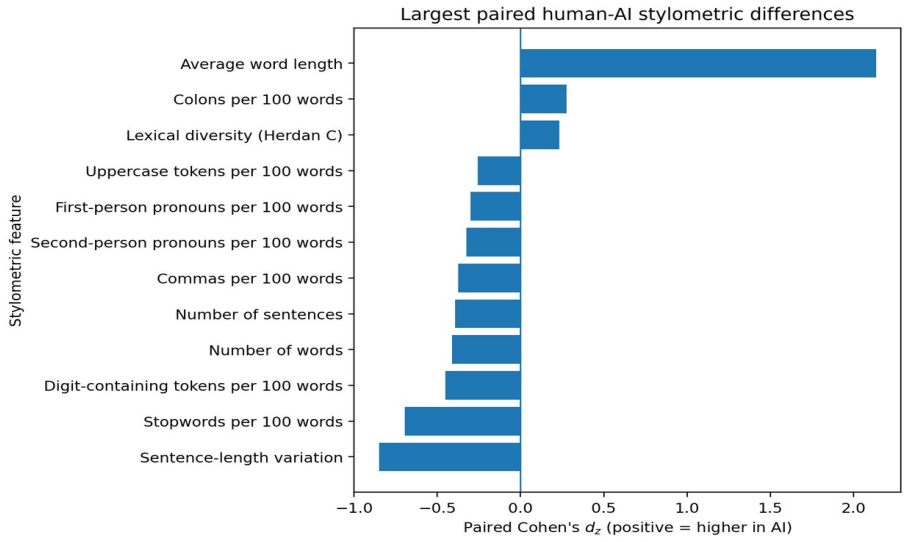


Figure B1. Largest paired human-AI stylometric differences after leading-Markdown normalisation.

Table B2. Paired stylometric effect sizes for all 19 features

Feature	Cohen's d_z
Average word length	2.1363
Sentence-length variation	-0.8513
Stopwords per 100 words	-0.6966
Digit-containing tokens per 100 words	-0.4533
Number of words	-0.4103
Number of sentences	-0.3947
Commas per 100 words	-0.3754
Second-person pronouns per 100 words	-0.3260
First-person pronouns per 100 words	-0.3007
Colons per 100 words	0.2747
Uppercase tokens per 100 words	-0.2571
Lexical diversity (Herdan C)	0.2344
Question marks per 100 words	-0.2239
Parentheses per 100 words	-0.1839
Modal verbs per 100 words	-0.1765
Quotation marks per 100 words	-0.1462
Semicolons per 100 words	-0.1422
Exclamation marks per 100 words	-0.0848
Average sentence length	-0.0279

Positive d_z values indicate higher values in AI-generated text and negative values indicate higher values in human-written text. Seventeen of the 19 features remained significant after Benjamini–Hochberg correction; average sentence length and semicolon rate were not significant.

B.3 Style-only logistic regression

Figure B2 reproduces the notebook's standardised coefficient plot for the 17-feature style-only logistic regression. The 17 features were standardised using training-set parameters (Appendix A.7); coefficients are conditional on the other standardised features.

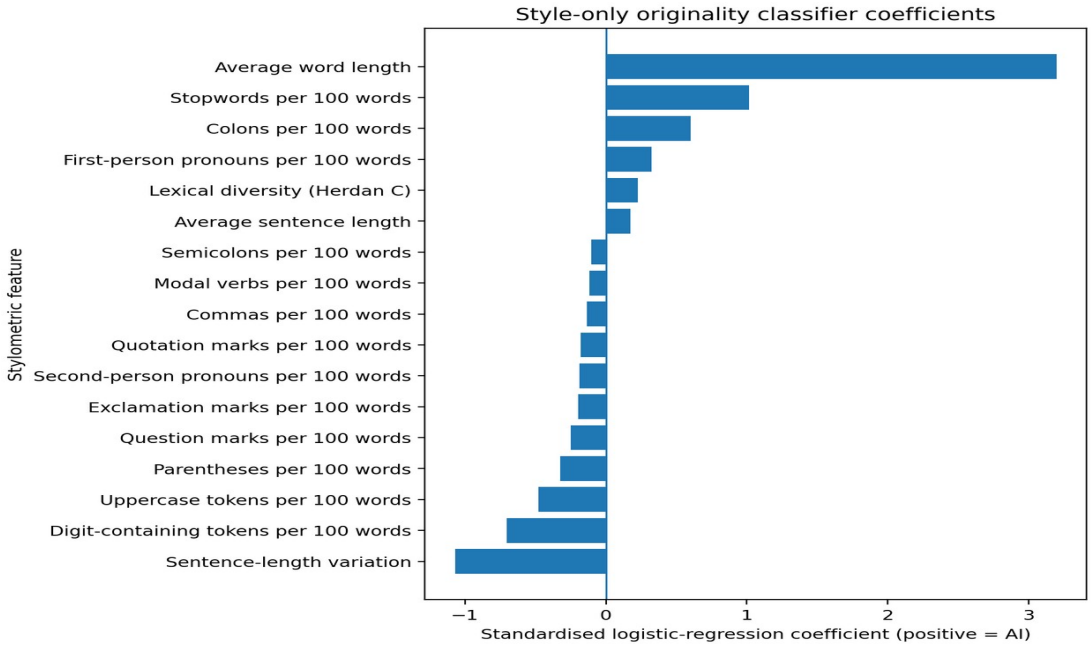


Figure B2. Standardised coefficients for the style-only originality classifier.